Edge-Guided Recurrent Convolutional Neural Network for Multitemporal Remote Sensing Image Building Change Detection

Beifang Bai, Wei Fu, Ting Lu, *Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Building change detection is a very important application in the field of remote sensing. Recently, deep learning (DL) has been introduced to solve the change detection task and achieved good performance, mainly due to the capability of automatically learning deep features. However, the lack of using prior knowledge (e.g., edge structure information) leads to inaccurate detection results, especially in the areas of building boundaries. To solve this problem, an end-to-end DL method for building change detection, named by edge-guided recurrent convolutional neural network (EGRCNN), is proposed in this article. The main idea is to incorporate both discriminative information and edge structure prior in one framework to improve change detection results, especially to generate more accurate building boundaries. First, a siamese convolutional neural network is trained to simultaneously extract primary multilevel features from multitemporal images. Then, a difference analysis module (DAM) is introduced to further produce discriminative features, which is constructed based on the basic long short-term memory module. Finally, both the discriminative features and the estimated edge structure information are jointly exploited to predict building change map. On one hand, the proposed DAM helps to enhance the discrimination between the changed and unchanged regions. On the other hand, the prior edge information is used to push the predicted changed buildings to preserve the original structure, which can further improve the accuracy of building change detection. Experimental results demonstrate that the performance of the proposed method outperforms several state-of-the-art approaches, in terms of objective metrics and visual comparison results.**

***Index Terms*— Building change detection, edge structure prior, multitemporal images, recurrent convolutional neural network (CNN).**

I. Introduction

BUILDING change detection aims to analyze multitemporal images in the same area, to monitor the dynamic change of buildings. It has been widely used in urban planning, illegal construction management, and land resource utilization [1]–[4]. With the rapid development of satellite imaging technology, a large amount of remote sensing (RS) images are available, which triggers the demand for intelligent change detection technology.

Previously, various traditional change detection methods [5]–[15] are proposed. These methods can be divided into three categories, i.e., image algebra-based method, image transformation-based method, and image classification-based method. The well-known change vector analysis (CVA) [5] is one classical change detection method based on image algebra. The CVA method calculates the change intensity and change direction between pixels in multitemporal images and sets thresholds to distinguish changed and unchanged regions. For the image transformation-based method, specific feature representations can be learned from the original image to better discriminate between the changed and unchanged pixels. For example, Celik [6] applied a principal component analysis to extract effective difference features from multitemporal images. Multivariate alteration detection [7] and iteratively reweighting multivariate alteration detection [8] realize change detection by maximizing the variance of image difference. Ertrk *et al.* [13] used sparse unmixing to provide pixel- and subpixel-level change information for hyperspectral data. Ding *et al.* [14] generated difference features by stacking multitemporal multiscale center-symmetric local binary pattern features and used the sparse reconstruction error to classify the difference features. In addition, many classification-based methods have been widely used. For instance, Bruzzone *et al.* [11] designed an ensemble of nonparametric multitemporal classifiers under the framework of the compound classification decision rule to detect changes. Deng *et al.* [12] combined the unsupervised classification and supervised classification into a hybrid classifier to detect land-use changes. Wu *et al.* [15] fused the classification probability map and the change probability map through Bayesian theory to obtain the final change map.

In recent years, deep learning (DL) has become a powerful tool in various computer vision tasks, including scene classification, semantic segmentation, and target detection. Due to that the DL can extract deep features to better represent image information, it has also been widely used in the RS image processing domain [16]–[30]. For example, Sun *et al.* [16] proposed one unified part-based convolutional neural network (CNN) for composite object detection in RS

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images. He *et al.* [19] proposed a lightweight network for cloud detection task, resulting in the boundary of the detected object is clear and the false-alarm rate is low. Hong *et al.* [20] used graph convolutional network and CNN for hyperspectral image classification, by effectively fusing different types of features, which significantly improved the overall classification performance. Hong *et al.* [22] developed a general multimodal DL framework, which introduced a new solution to generate good RS images classification results of complex scenes. As one classical network, CNN is introduced in some works for detecting changes in RS images, which can produce good results. For example, Peng *et al.* [31] first stacked the multitemporal images as the input of the UNet++ [32] to generate multilevel feature maps, which are used for predicting multiple change maps. Then, the multiside-output fusion strategy (MSOF) is introduced to combine the primary multiple change maps to generate the final prediction results. In addition, Peng *et al.* [33] embedded the attention mechanism into UNet++ to improve the performance of the network. The well-known siamese network is originally proposed for measuring the similarity between image pairs, which has also been used for change detection. For example, Chen and Shi [34] used two branches of the siamese network to simultaneously extract the features of the multitemporal images, and then, the Euclidean distance is exploited to measure the difference between the features. Moreover, Jiang *et al.* [35] combined the siamese network and various attention mechanisms to enhance the discrimination capability of changed and unchanged areas.

In addition to the CNN-based change detection methods, recurrent neural network (RNN) is also found to be effective in processing change detection tasks. As an important technology of DL, the RNN has a promising capability in processing sequence data. By regarding multitemporal images as sequence data, the RNN can be used for change detection. Mou *et al.* [36] first used the siamese network to extract the features of multitemporal images and then used a special RNN architecture [i.e., long short-term memory (LSTM)] to mine the temporal dependence of these features. Chen *et al.* [37] constructed a multiple-layer RNN by stacking LSTM units to extract change information. In addition, some methods attempt to take advantage of both the DL methods and traditional methods, in order to improve the change detection performance. For example, Du *et al.* [38] proposed an unsupervised change detection method by integrating the deep network and slow feature analysis. Saha *et al.* [39] combined the deep network and the CVA to generate the deep change feature vector and analyze its magnitude to identify changed pixels.

Compared with the traditional machine learning methods, the DL-based change detection methods present a more promising performance. However, it is still a challenge to accurately classify the edge pixels of changed buildings due to the complex scene. For example, factors of dense distribution of buildings, the effect of shadow, and the similarity between buildings and roads will make the change detection methods dependent on the discriminative features that are not effective. In order to refine the inaccurate edges, some works [30], [31], [40] adopt a multilevel feature fusion scheme. By incorporating shallow features containing spatial information with the deep features representing abstract information, these methods can achieve improved detection performance; meanwhile, more accurate object edges can be obtained. However, these methods ignore the prior information based on the edge structure of the target. As one typical prior, the edge structure information is considered here due to the effectiveness and simple operation. First, for a building, it is a relatively fixed-shaped object that appears as a polygon. Edge information can not only reflect the geometric shape of the building but also reflect the location information of the building. Exploiting edge information effectively will guide the building change detection process to generate a better prediction result, especially around the boundary areas. Second, two tasks of change detection and edge estimation are closely related, which can be jointly trained in one united framework to improve the robustness of DL network. Third, the edge detection operator (e.g., the canny operator) can quickly and accurately extract the edge prior information based on the labeled training sample set. Based on the above consideration, a new method called edge-guided recurrent convolutional neural network (EGRCNN) is proposed in this article. In brief, the edge structure based on prior is introduced to correct the wrongly classified edge pixels. First, preliminary multiscale features are extracted from multitemporal images via using the siamese network. Then, a difference analysis module (DAM) is proposed for difference information mining from the preliminary features. Finally, we developed an edge guidance module (EGM) to integrate difference information and structure information, to generate the estimated changed buildings. The main contributions are concluded as follows.

1) In this work, a novel end-to-end building change detection DL framework is introduced. Both the discriminative information and edge structure prior information are integrated into one DL framework to improve the building change detection results, especially to generate better edges of detected buildings.
2) In order to accurately distinguish the buildings from the background, the DAM is proposed to embed an adaptive weighting mechanism on the basis of the LSTM structure. The greater the distance between the preliminary feature pairs is, the higher weight will be assigned to the result of the LSTM, and vice versa. By this way, the difference information between multitemporal images can be effectively enhanced.
3) Aiming at obtaining buildings with fine edges, an EGM and a deep edge-guided loss function are designed for final change map estimation. First, an EGM is developed to estimate both the change detection probability and edge prediction probability. Then, a deep edge-guided loss function is designed by jointly exploiting both probabilities. Along with the training process, the constraints of the two-term loss function forces the final predicted changed building to preserve its original boundaries.

The remainder of this article is organized as follows. Section II discusses the related work. Section III introduces the details of the proposed network. Experimental results and analysis on the LEVIR-CD dataset and the WHU building change

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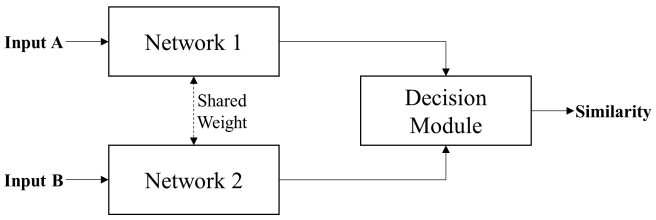


Fig. 1. Architecture of siamese network.

detection dataset are provided in Section IV. Finally, Section V summarizes our work.

II. Related Work

A. Siamese Network

As shown in Fig. 1, the siamese network has two branches, which have exactly the same network structure and the same weights. The two branches accept different inputs and are combined by a decision module, which can measure the similarity of the two inputs. The purpose of change detection is to distinguish the changed pixels from the unchanged pixels in the multitemporal images. From this aspect, the siamese network is a good candidate solution for change detection tasks. First, the two branches of the siamese network are used to extract the features from multitemporal images. Then, the change maps are generated by measuring the similarity between the pair of features. Daudt *et al.* [41] compared the early fusion method with the siamese network. The difference with the siamese network is that the early fusion method is to first concatenate multitemporal images as one image and then input it into the deep network to realize change detection. The effectiveness of the siamese network is proven by comparison results.

B. Long Short-Term Memory

Compared with the traditional RNN, the LSTM adds a cell state and three gates. By this way, a better performance can be obtained by the LSTM in terms of analyzing the correlation of sequence data, which has been demonstrated in some related works [36], [37]. The three gates of the LSTM network are forget gate, input gate, and output gate. The specific operation of the LSTM is as follows.

First, the LSTM network updates the cell state c_t by selectively discarding previous information of the cell state c_{t-1} through the forget gate f_t and adding new information of the cell state $\tilde{c}_t$ through the input gate i_t

$$c_t = i_t \odot \tilde{c}_t + f_t \odot c_{t-1} \tag{1}$$

$$\tilde{c}_t = \tanh(W_{ci} x_t + W_{ch} h_{t-1}) \tag{2}$$

where $\odot$ is an elementwise operation that multiplying the corresponding elements, $\tanh(\cdot)$ is the hyperbolic tangent function, and W represents the weight matrix, whose subscripts have special meanings. For example, W_{ci} is the memory-input weight matrix, W_{ch} is the memory-hidden weight matrix, x_t is the new input, and h_{t-1} is hidden state. The input gate i_t and the forget gate f_t are, respectively, expressed as

$$f_t = \sigma\left(W_{fi} x_t + W_{fh} h_{t-1} + W_{fc} c_{t-1}\right) \tag{3}$$

$$i_t = \sigma\left(W_{ii} x_t + W_{ih} h_{t-1} + W_{ic} c_{t-1}\right) \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid function.

Then, the output gate o_t controls the output information of the cell state c_t, and h_t represents the final output of the LSTM network

$$h_t = o_t \odot \tanh(c_t). \tag{5}$$

Specifically, the output gate o_t is computed as follows:

$$o_t = \sigma(W_{oi} x_t + W_{oh} h_{t-1} + W_{oc} c_t). \tag{6}$$

III. Proposed Method

As shown in Fig. 2, the proposed EGRCNN method mainly consists of three steps: multilevel features extraction, difference information extraction, and edge-guided change map estimation. For the first step, a pair of images with size of 256×256 is input to the siamese network to extract preliminary multilevel features of the original image. Then, a DAM is introduced to extract the difference information from the multilevel feature pairs. Finally, an EGM is proposed to integrate both the difference information and edge structure information to estimate the final change map. The detailed descriptions are given in the following.

A. Multilevel Features Extraction

Buildings in RS images always have different sizes and shapes, and one-level feature cannot well represent the characters of changed buildings. Based on this consideration, the multilevel features are extracted, which contains both the deep semantic features and shallow appearance features. Here, the siamese network is introduced to achieve this feature extraction task due to the particular network structure. In specific, the siamese network has two branches, which shares the same network structure and weights. Since the input data of the change detection contain two images, the two-branch structure of the siamese network is particularly suitable for this task. Besides, the siamese network will effectively extract deep abstract features, which can well characterize each of the multitemporal images. Moreover, the weight sharing between two branches can map multitemporal images to the same feature space, making the extracted features more distinguishable. The U-Net [42] is chosen as the backbone of the siamese network because of its lightweight structure and excellent performance. Both branches of the siamese network have the same structure as the encoder of U-Net. Since each branch contains four max-pooling layers, five pairs of different levels features are extracted. These feature pairs are expressed as F_{1i} and F_{2i}, where $i \in \{1, 2, 3, 4, 5\}$ represents the different levels. In this way, each sample pair in multitemporal images can be represented by multilevel features.

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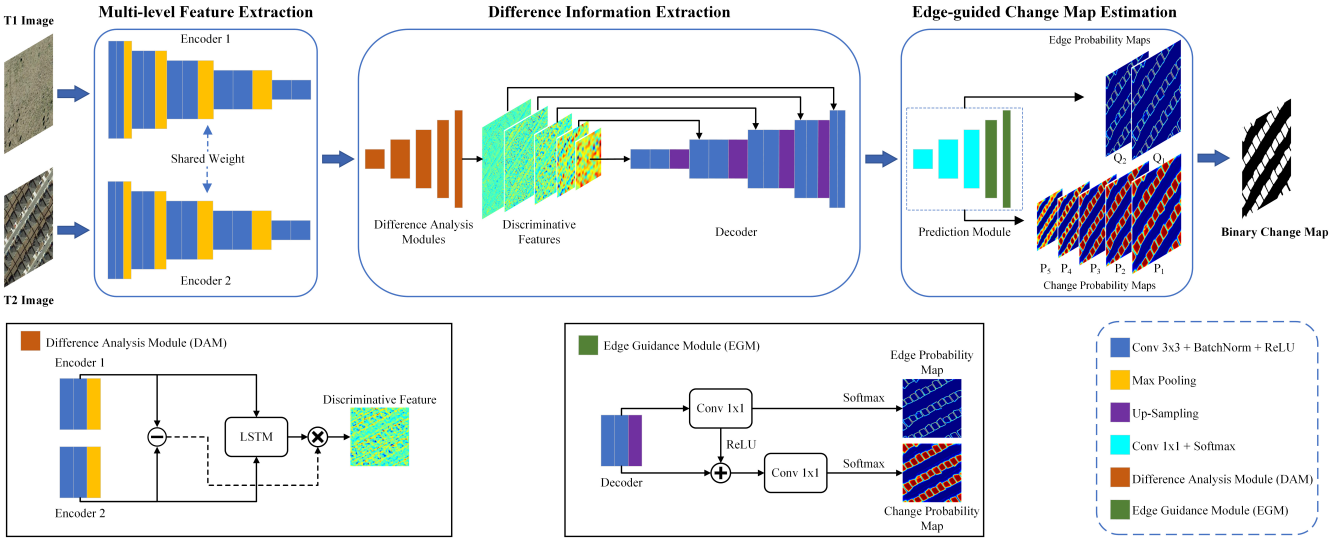


Fig. 2. Flowchart of the proposed EGRCNN building change detection method.

B. Difference Information Extraction

To further excavate more accurate difference information from multilevel features, a DAM is constructed via analyzing the correlation between feature pairs in a coarse-to-fine manner.

In the "coarse" stage, the LSTM is used to initially measure the similarity between features F_{1i} and F_{2i}, leading to the generation of preliminary discriminative features L_i. The reasons why the LSTM is introduced here are explained next. First, as a kind of RNN, the LSTM is good at modeling the temporal correlation between sequence data. Since multitemporal images also belong to a kind of sequence data, the LSTM is exploited to analyze the temporal correlation between the input image pairs. Besides, affected by some natural factors, e.g., illumination conditions and season changes, the multitemporal images acquired at different times always show a great spatial–spectral difference. In this situation, the linear difference analysis methods are difficult to achieve accurate change detection results. For example, pseudo-changes, such as changes of vegetation color in different seasons, easily lead to inaccurate detection results. The LSTM is a nonlinear difference analysis method, which helps to enhance the robustness of the network.

Then, in the "fine" stage, an adaptive weighting mechanism is designed to enhance difference information, where the weights are adjusted adaptively according to L_i. When a larger distance is between preliminary feature pairs, a higher weight will be assigned and vice versa. This is reasonable because a larger distance means a higher probability of the sample pair belonging to changed pixels. In this way, the DAM can effectively enhance the difference information. Specifically, the multilevel discriminative features D_i can be formulated as follows:

$$D_i = |F_{1i} - F_{2i}| \odot L_i. \tag{7}$$

Here, the high-level discriminative features represent semantic information, which can help in locating changed areas. The low-level discriminative features contain detailed information, which can better reconstruct the spatial structure of the changed buildings. Then, all the multilevel discriminative features are combined by a decoder process. Considering the symmetrical structure of the encoder and decoder, the decoder in EGRCNN contains four upsampling layers. In the decoding process, the deep features after upsampling and the shallow features of the corresponding resolution are concatenated through the skip connection. In this way, a more compact difference feature can be obtained, where all the complementary information within multilevel discriminative features is maintained.

C. Edge-Guided Change Map Estimation

The first two steps of multilevel feature extraction and difference information extraction are designed to better extract discriminative features between the changed and unchanged regions. To further improve the performance of detected boundaries of changed buildings, the next step is the introduced edge-guided change map estimation. On one hand, edge information can not only reflect the geometric shape of the building but also reflect the location information of the building. Extracting edge information effectively will guide the building change detection process to generate a better prediction result, especially around the boundary areas. On the other hand, better change detection results will help to improve the edge prediction results since the edge information is extracted from the change estimation maps here, that is, the two tasks of change detection and edge prediction are closely correlated and can promote each other in terms of detection performance. Based on these considerations, an EGM is introduced here, where the edge estimation task is integrated with the building change prediction task in one DL framework.

As shown in Fig. 2, the introduced EGM consists of two branches. The first branch estimates the edge information from the output of the decoder through the 1×1 convolution block. The other branch combines the edge information with

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the output of the decoder and then estimates the change information through the 1×1 convolution block. At last, both the edge information and change information are through softmax layers to obtain the change probability maps and edge probability maps. The EGM can not only predict changed buildings and their edges but also integrate edge features directly into discriminative features to further improve the quality of the prediction results. In order to improve the performance of the proposed EGRCNN, deep supervision is introduced to additionally supervise the intermediate layers of the network. In the training process, the intermediate layers directly receive feedback from the ground truth change maps of different spatial resolutions, making it not only dependent on the gradients backpropagated from the output layer. Therefore, the network can quickly converge and produce more representative discriminative features. In our framework, the EGM predicts the output of the first two layers of the decoder to obtain the change probability maps and the edge probability maps. Since the output of the other three layers has low spatial resolution and lack of structural information, it is difficult to predict the edges of the buildings. The output of these layers only uses the 1×1 convolution block to predict the change probability maps. As a result, the network will output five change probability maps and two edge probability maps.

In the training process, a deep edge-guided loss function is designed to train the whole network, which consists of change detection loss ($\mathcal{L}_1$) and edge detection loss ($\mathcal{L}_2$). Considering the fact that unchanged areas are much larger than changed areas, we adopt the focal loss [43] to solve the problem of imbalanced positive and negative training samples for change detection. The focal loss can be expressed as

$$\mathcal{L}_1 = -\frac{1}{N}\sum_{n=1}^{N}[\alpha(1-\hat{y}_n)^{\gamma} y_n \log \hat{y}_n + (1-\alpha)\hat{y}_n^{\gamma}(1-y_n)\log(1-\hat{y}_n)] \quad (8)$$

where N is the number of samples, $\hat{y}_n$ is the predicted probability value, and y_n is the true value. By adjusting the values of weighting factor α and focusing parameter γ to reduce the weight of negative samples, the network pays more attention to positive samples in the training process.

To define edge detection loss, the mean squared error (MSE) loss function is used, which can be described as follows:

$$\mathcal{L}_2 = \frac{1}{N}\sum_{n=1}^{N}(z_n - \hat{z}_n)^2 \quad (9)$$

where z_n and $\hat{z}_n$, respectively, represent the true value and the predicted probability value in the edge detection. Finally, the deep edge-guided loss is shown as follows:

$$\mathcal{L} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2. \quad (10)$$

By defining the loss function as in (10), the restriction of discrimination information and edge structure prior is considered jointly. Besides, the two items in (10) are balanced by a predefined weight, λ_1 and λ_2. Along with the optimization of the loss function, the final predicted changed buildings will have higher accuracy while maintaining better boundaries.

In the testing process, we only use the change probability map with the highest spatial resolution (P1 in Fig. 2) and set the threshold τ to transform it into the final binary change map. The details of the threshold method can be described by the following formula:

$$M_{a,b} = \begin{cases} 1, & P_{a,b} \geq \tau \\ 0, & \text{else} \end{cases} \quad (11)$$

where $P_{a,b}$ denotes the change probability at pixel (a, b) and $M_{a,b}$ indicates if the current pixel belongs to a changed building. In specific, if the probability $P_{a,b}$ has a higher value of τ, it will be regarded as a changed pixel, i.e., $M_{a,b} = 1$, and vice versa.

IV. Experimental Results and Discussion

In this section, the performance of the proposed EGRCNN method[1] is evaluated experimentally on two building change detection datasets, including LEVIR-CD dataset[2] [34] and WHU building change detection dataset[3] [44]. To demonstrate the effectiveness of the proposed method, five methods based on DL are compared numerically and visually. These methods include ChangeNet [45], spatial–temporal attention neural network (STANet) [34], UNet-LSTM [46], MSOF [31], and feature difference convolutional neural network (FDCNN) [40]. In terms of objective metrics to measure the change detection performance, five commonly used metrics, such as precision (P), recall (R), F1 score (F1), overall accuracy (OA), and Kappa coefficient (KC), are calculated. In addition, the visual comparison results of the change detection results are also shown in the following.

A. Data Description

LEVIR-CD dataset and WHU building change detection dataset are used to evaluate the performance of the proposed method. Fig. 3 shows the examples from the two datasets. In particular, Fig. 3(a) and (b) shows the image pairs from multitemporal images that are acquired in the same area at different times. Fig. 3(c) shows the related ground truth of building change maps, where the white and black values indicate changed and unchanged pixels. The edge detection results in Fig. 3(d) are generated by conducting the canny edge detection operator on Fig. 3(c).

1) LEVIR-CD Dataset: The LEVIR-CD dataset contains 445 training datasets, 64 validation datasets, and 128 test datasets, and each dataset contains a pair of multitemporal images with size of 1024×1024. These images are from 20 different areas in several cities in TX, USA, with a spatial resolution of 0.5 m. The LEVIR-CD dataset contains a total of 31 333 changed buildings. Due to the large size of the image, each image in the LEVIR-CD dataset is cropped into 16 nonoverlapping images with size of 256×256. By this way, we generate a dataset consisting of 7120 training sets, 1024 validation sets, and 2024 test sets.

[1] https://github.com/luting-hnu/EGRCNN
[2] https://justchenhao.github.io/LEVIR/
[3] https://study.rsgis.whu.edu.cn/pages/download/

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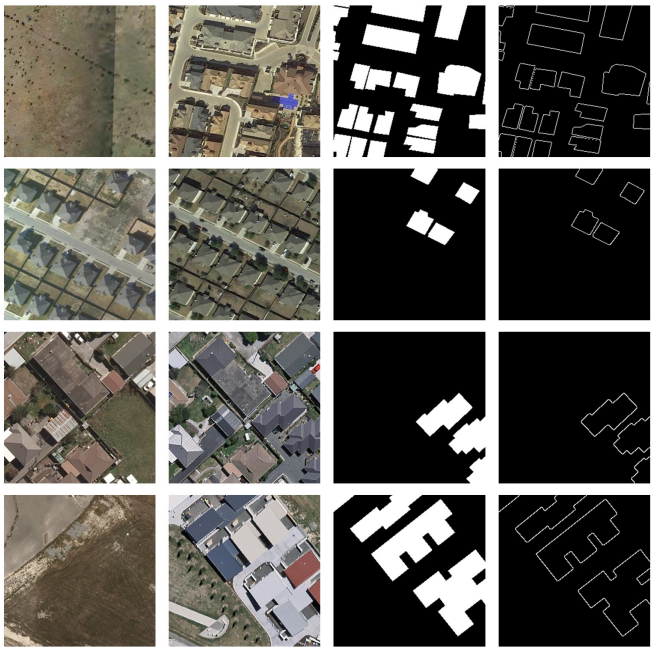


Fig. 3. First and second rows are cropped examples from the LEVIR-CD dataset. The third and fourth rows are cropped examples from the WHU building change detection dataset. (a) T1 image. (b) T2 image. (c) Ground truth change map. (d) Building edge map.

2) WHU Building Change Detection Dataset: The WHU building change detection dataset covers the area of the 6.3-magnitude earthquake that struck Christchurch, New Zealand, in February 2011. The dataset contains a pair of images at the same area taken in 2012 and 2016. The size of each image is 32 507 × 15 354 pixels, which is then cropped into multiple nonoverlapping image blocks with size of 256 × 256 for the following experiments. The image blocks are divided into training sets, validation sets, and test sets with the corresponding ratio of 7:1:2, that is, we have 5205 training sets, 743 validation sets, and 1486 test sets.

B. Experimental Setting

To measure the performance of the proposed network, we utilize five commonly used evaluation metrics: precision (P), recall (R), F1 score (F1), overall accuracy (OA), and KC. The formulas of these evaluation metrics are as follows:

$$\mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{12}$$

$$\mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{13}$$

$$\mathrm{F1} = \frac{2\mathrm{PR}}{\mathrm{P}+\mathrm{R}} \tag{14}$$

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{15}$$

$$P_e = \frac{(\mathrm{TN}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})+(\mathrm{FP}+\mathrm{TP})(\mathrm{FN}+\mathrm{TP})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \tag{16}$$

$$\mathrm{KC} = \frac{\mathrm{OA}-\mathrm{P_e}}{1-\mathrm{P_e}} \tag{17}$$

where TP is the number of true positives, FP is the number of false positives, TN is the number of true negatives, and FN is the number of false negatives.

In our experiments, the proposed EGRCNN method is implemented with Pytorch. We use the Adam optimizer and learning rate decay strategy to train this network. The input image has data augmentation, including random graying and random adjustment of brightness and contrast. The batch size is set to be 12 and the base learning rate is set to be $1e^{-}4$. When the F1 score of the validation set does not increase for 15 consecutive epochs, the learning rate drops to 0.1 times the current learning rate. The entire training process includes 100 epochs and the threshold τ is set to be 0.5. The parameters α and γ in the focal loss are set to 0.25 and 2, respectively. The weight ratio of the deep edge-guided loss function is set to $\lambda_1 : \lambda_2 = 1 : 10$ to accelerate the learning progress of the edge. The parameters of the methods in the comparison experiment are set as the default parameters in the original papers. All experiments are repeated ten times to obtain the mean and standard variances of precision, recall, F1, OA, and KC.

C. Parameter Discussion

For the proposed EGRCNN method, the important parameter of the number of EGMs needs to be predefined. To validate the effectiveness of EGM, we designed experiments EGRCNN_n for comparison where n is the total number of EGMs, $n \in \{0, 1, 2, 3, 4, 5\}$. EGRCNN_0 means that EGM is skipped, i.e., the edge information is not used. EGRCNN_1 means that only the output of the first layer in the decoder uses EGM to predict the change probability map and edge probability map. The output of the remaining four layers in the decoder uses the 1 × 1 convolution block to predict the change probability maps. EGRCNN_2 uses EGMs for the output of the first two layers of the decoder. By analogy, EGRCNN_5 means that the output of all layers in the decoder uses EGMs to predict the probability maps, which contains five change probability maps and five edge probability maps. Fig. 4 shows the influence of n on the building change detection performance, in terms of five objective metrics, i.e., precision, recall, F1, OA, and KC. It can be seen that along with the increase of n, the values of F1, OA, and KC all show a trend of first increasing and then decreasing. In specific, the highest values for these four metrics can be obtained when $n = 2$. This can be explained as follows. The introduction of edge information makes the edge of the building clearer and the structure more complete. However, as the downsampling of the original images, e.g., $n \geq 3$, the estimated edge information becomes inaccurate, leading to the decrease of change detection results. From Fig. 4, the best recall can be obtained when $n = 5$, while a competitive performance is obtained when n is set to be values between 0 and 4. With the introduction of inaccurate structural information, the EGRCNN_5 is more inclined to predict the pixels around edges to be changed areas. As a result, a high recall will be obtained when $n = 5$, at the cost of a decrease of precision. Based on the above analysis, to generate the optimal performance of building change detection, the value of n is set to 2 in the following experiments.

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TABLE I

COMPARISON OF QUANTITATIVE INDEXES OF ABLATION STUDIES ON THE LEVIR-CD DATASET. THE W/O EGM MEANS THE EGRCNN METHOD WITHOUT THE EGM AND THE W/O DAM MEANS THE EGRCNN WITHOUT THE DAM

Methods	P(%)	R(%)	F1(%)	OA(%)	KC(%)
w/o DAM	87.37 ± 0.72	$\mathbf{91.59 \pm 0.74}$	89.43 ± 0.22	98.90 ± 0.03	88.85 ± 0.24
w/o EGM	87.26 ± 0.47	91.45 ± 0.69	89.30 ± 0.18	98.89 ± 0.02	88.72 ± 0.19
EGRCNN	$\mathbf{87.82 \pm 0.50}$	91.47 ± 0.50	$\mathbf{89.61 \pm 0.16}$	$\mathbf{98.92 \pm 0.02}$	$\mathbf{89.04 \pm 0.17}$

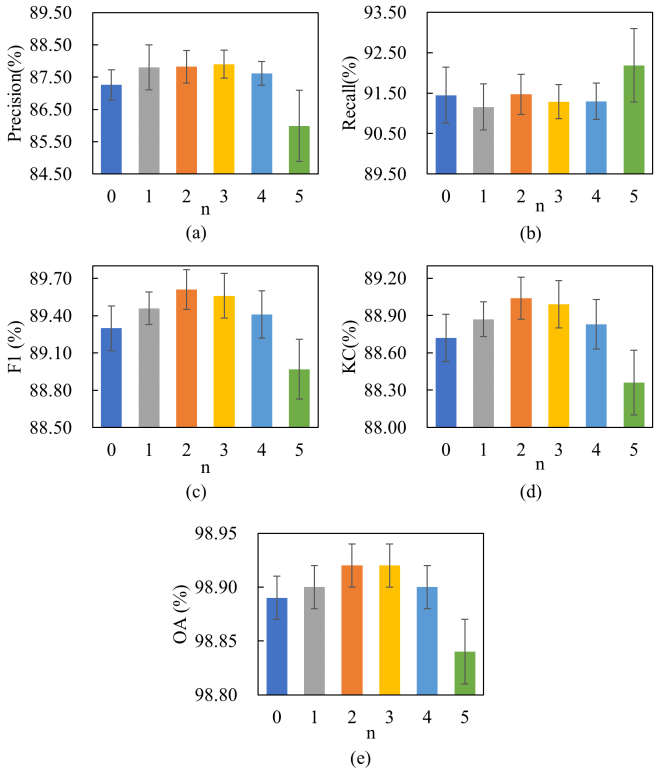


Fig. 4. Influence of the parameter n (different number of EGMs) on the change detection performance of the proposed method in terms of (a) precision, (b) recall, (c) F1, (d) OA, and (e) KC.

D. Ablation Studies

In order to validate the effectiveness of each component of the proposed network, we designed ablation experiments on the LEVIR-CD dataset, by conducting the experiments neglecting different modules in the proposed framework. The results are shown in Table I, and the best results in this table have been marked in bold.

In specific, to study the impact of DAM on the network, we designed a new method with LSTM instead of DAM. The method is named by w/o (i.e., without) DAM, and the result is shown in the first row of Table I. It can be observed that the EGRCNN achieves higher values on three comprehensive evaluation metrics of F1, OA, and KC, demonstrating the effectiveness of the introduced DAM. Though the recall of w/o DAM is 0.12% higher when compared with the EGRCNN, the precision is 0.45% lower. One main reason for this is that the metric of recall and precision always presents a negative correlation. It is difficult to obtain high precision and recall at the same time. To further illustrate the superiority of introducing the DAM, we give some visual comparison results here, as shown in Fig. 5. In the first row of Fig. 5, the w/o DAM method misidentifies the playground as a changed building. By contrast, EGRCNN with the DAM can strengthen the anti-interference ability of the network and better discriminate between buildings and other objects that have similar spectral information. Furthermore, EGRCNN can detect the changed buildings, even if they are obscured by trees (row 2 in Fig. 5). This is because the DAM implements the change enhancement function on the basis of LSTM, leading to the improvement of overall detection performance.

To validate the effectiveness of EGM, we use 1×1 convolution blocks instead of EGM, named by the w/o (i.e., without) EGM. The results are shown in the second row of Table I. Compared with the EGRCNN, the five metrics by the w/o EGM method have different degrees of decrease, i.e., precision (0.56%), recall (0.02%), F1 (0.31%), OA (0.03%), and KC (0.32%). The numerical comparison results demonstrate that the proposed EGRCNN has a better performance. As can be seen in Fig. 6, the edges of the buildings in the change maps generated by w/o EGM are blurred, and even two buildings that are relatively close are identified as one. In contrast, EGRCNN takes advantage of edge information to make the edges of buildings in the change maps more complete. Meanwhile, those densely distributed buildings can also be accurately distinguished.

E. Performance Comparison

In this section, the performance of the proposed framework is compared with five recently proposed change detection methods, including the ChangeNet, STANet, UNet-LSTM, MSOF, and FDCNN. ChangeNet is a change detection network based on ResNet50, which realizes the monitoring of changed areas by fusing change information of different scales. The STANet is constructed based on ResNet18, which uses the self-attention mechanism to model the spatial and temporal correlations. The UNet-LSTM explores both the UNet and LSTM for change analysis. The MSOF based on UNet++ uses the MSOF strategy to combine different semantic levels of change maps. The FDCNN introduces the siamese network to generate multiscale and multidepth difference features for change detection. Note that in order to make a fair comparison, FDCNN does not use the RS image scene classification dataset for auxiliary training.

In Table II, the quantitative analysis results of all the compared methods on the LEVIR-CD dataset are listed. It can be seen that the EGRCNN achieves the best results in terms of precision (87.82%), F1 (89.61%), OA (98.92%), and KC (89.04%). In terms of the metric of recall, the EGRCNN has

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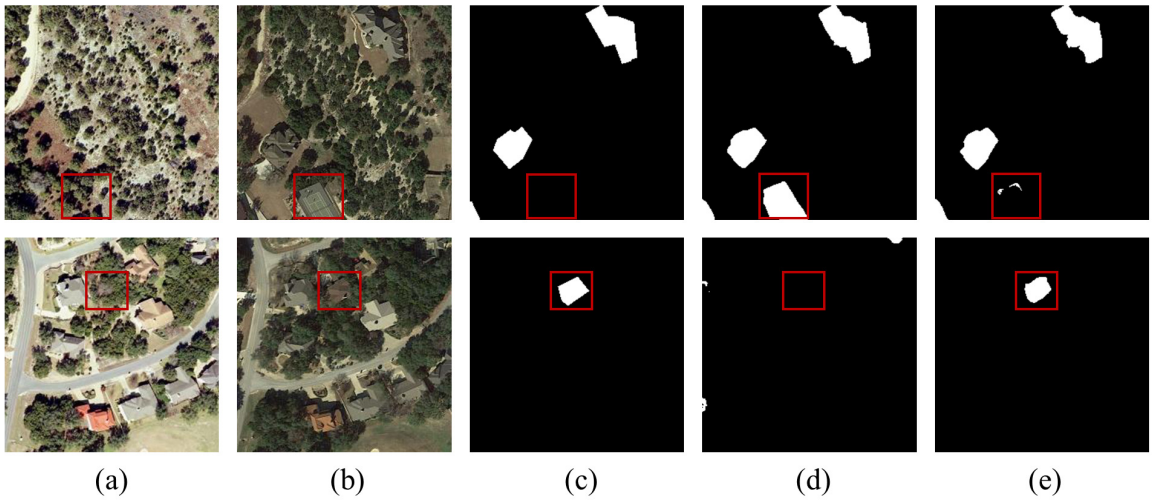


Fig. 5. Examples of ablation experiments on the LEVIR-CD test dataset. (a) T1 image. (b) T2 image. (c) Ground truth change map. (d) Change map produced by w/o DAM (i.e., EGRCNN without DAM). (e) Change map produced by EGRCNN.

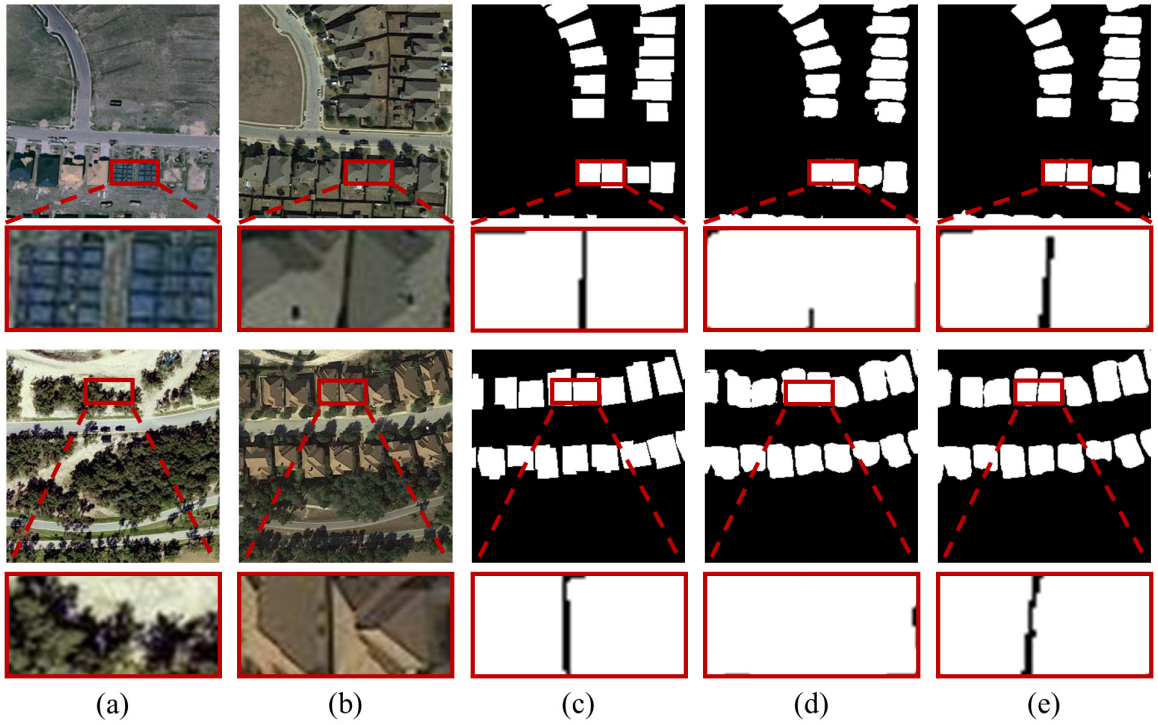


Fig. 6. Examples of ablation experiments on the LEVIR-CD test dataset. (a) T1 image. (b) T2 image. (c) Ground truth change map. (d) Change map produced by the w/o EGM (i.e., EGRCNN without EGM). (e) Change map produced by EGRCNN.

TABLE II
COMPARISON OF QUANTITATIVE METRICS FOR LEVIR-CD BUILDING CHANGE DETECTION DATASET

Methods	P(%)	R(%)	F1(%)	OA(%)	KC(%)
ChangeNet [45]	78.85 ± 0.76	83.32 ± 0.58	81.02 ± 0.30	98.01 ± 0.04	77.97 ± 0.32
STANet [34]	82.65 ± 1.88	89.66 ± 1.06	86.00 ± 1.06	98.51 ± 0.13	85.44 ± 1.07
UNet-LSTM [46]	85.45 ± 0.53	$\mathbf{91.65 \pm 0.53}$	88.43 ± 0.16	98.78 ± 0.02	87.80 ± 0.17
MSOF [31]	85.55 ± 0.60	89.22 ± 0.77	87.34 ± 0.12	98.68 ± 0.01	86.65 ± 0.13
FDCNN [40]	83.41 ± 1.18	90.68 ± 0.67	86.89 ± 0.56	98.60 ± 0.07	86.15 ± 0.59
EGRCNN	$\mathbf{87.82 \pm 0.50}$	91.47 ± 0.50	$\mathbf{89.61 \pm 0.16}$	$\mathbf{98.92 \pm 0.02}$	$\mathbf{89.04 \pm 0.17}$

a slightly lower recall value than the UNet-LSTM, but the detection precision by the EGRCNN method is 2.37% higher than the compared methods. One main reason for the slightly lower recall by the proposed method is the existing mislabeled samples in the ground truth. For example, the gap between buildings in the area within the red box is mislabeled as the

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TABLE III
COMPARISON OF QUANTITATIVE METRICS FOR WHU BUILDING CHANGE DETECTION DATASET

Methods	P(%)	R(%)	F1(%)	OA(%)	KC(%)
ChangeNet [45]	83.32 ± 3.08	73.75 ± 2.20	78.18 ± 0.98	98.19 ± 0.10	77.24 ± 1.02
STANet [34]	87.65 ± 1.64	85.15 ± 1.42	86.36 ± 0.46	98.82 ± 0.04	85.82 ± 0.49
UNet-LSTM [46]	86.72 ± 1.42	87.72 ± 0.65	87.21 ± 0.59	98.87 ± 0.06	86.62 ± 0.62
MSOF [31]	88.49 ± 1.35	84.87 ± 2.11	86.61 ± 0.60	98.84 ± 0.04	86.01 ± 0.62
FDCNN [40]	82.54 ± 1.22	87.33 ± 1.42	85.24 ± 0.59	98.63 ± 0.06	84.53 ± 0.62
EGRCNN	$\mathbf{89.90 \pm 0.98}$	$\mathbf{87.95 \pm 1.24}$	$\mathbf{88.90 \pm 0.43}$	$\mathbf{99.03 \pm 0.04}$	$\mathbf{88.40 \pm 0.45}$

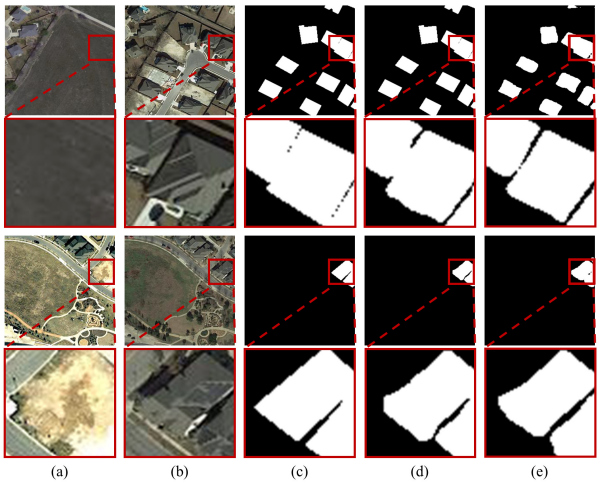


Fig. 7. Examples from the LEVIR-CD dataset, unchanged pixels are mislabeled as changed pixels. (a) T1 image. (b) T2 image. (c) Ground truth change map. (d) Change map produced by UNet-LSTM. (e) Change map produced by EGRCNN.

changing area in Fig. 7(c). Visually, it can be observed that the change maps produced by UNet-LSTM [see Fig. 7(d)] do not correctly classify these mislabeled regions, while the proposed EGRCNN [see Fig. 7(e)] does it. Thus, a higher FN (the number of changed pixels predicted to be unchanged pixels) value will be generated by the EGRCNN, leading to a decrease in recall. In addition, the metric of F1, OA, and KC can measure the algorithm performance from different aspects, indicating that a better detection performance is occupied by the proposed method. Among all the compared methods, the stability of the EGRCNN is also excellent. The standard deviations of precision and recall are 0.5%, and the standard deviations of F1, OA, and KC are 0.16%, 0.02%, and 0.17%, respectively. This is because the edge prior information improves the ability of the proposed method to recognize buildings, thereby enhancing the robustness of the algorithm.

Moreover, several examples of visual comparison results are given in Fig. 8, where there are six pairs of multitemporal images from the LEVIR-CD dataset. Among them, the first, second, and third rows show three examples of changed buildings with large size, and the fourth, fifth, and sixth rows give examples with dense buildings of smaller size. Due to the effect of shadow, the detected changed areas by the other five methods have inaccurate results especially around the edges, as can be seen from the first three rows. In contrast, the detection results by the EGRCNN can better match the structures in ground truth change maps. Another three examples (fourth, fifth, and sixth rows) show a bigger challenge because of the dense and small buildings, and it is clear that adjoining buildings are easily detected as one whole by the compared methods. Besides, different degrees of missed building detection results appear in especially the methods of ChangeNet, STANet, and MSOF. By introducing the structure prior, the EGRCNN can greatly improve the results of distinguishing between adjacent buildings, leading to more accurate detection results with more clear boundaries. In consistent with the numerical results in Table II, the visual comparison results also demonstrate the performance superiority of the proposed EGRCNN method.

Table III lists the quantitative analysis results on the WHU building change detection dataset, and Fig. 9 shows several change detection results of different methods on this dataset. From the table, it can be observed that the proposed EGRCNN method generates the best results in terms of five objective metrics. Among the compared method, the ChangeNet gets the most unsatisfied results, with the precision of 83.22%, recall of 73.75%, F1 of 78.18%, OA of 98.19%, and KC of 77.24%. In Fig. 9(d), many buildings are not detected by ChangeNet and the edges of the buildings are blurred. The precision of FDCNN is 82.541%, recall is 87.73%, F1 is 85.24%, OA is 98.63%, and KC is 84.53%. In Fig. 9(h), FDCNN fails to detect some of the changed areas and misdetects some of the unchanged areas to changed areas. The performance of STANet and MSOF is better than FDCNN, with the precision of 87.65% and 88.49%, recall of 85.15% and 84.87%, F1 of 86.36% and 86.61%, OA of 98.82% and 98.84%, and KC of 85.82% and 86.01%. As shown in Fig. 9(e) and (g), it is difficult to detect buildings in areas with complex environments and the internal compactness of buildings is low. The effect of UNet-LSTM is slightly better than MSOF with the precision of 86.72%, recall of 87.72%, F1 of 87.21%, OA of 98.87%, and KC of 86.62%. However, the edges of the buildings are incomplete in Fig. 9(f). In Fig. 9(i), the structure of the buildings detected by EGRCNN is complete and has the highest precision, recall, F1, OA, and KC, which are 89.90%, 87.95%, 88.90%, 99.03%, and 88.40%, respectively. In addition, the standard deviation of each metric of EGRCNN is also the smallest, which proves its stability. In summary, EGRCNN also achieved the best performance on the WHU building change detection dataset. This fully proves the effectiveness of the proposed method and the robustness to different datasets.

In order to further verify the overall performance of the proposed method, we draw the precision–recall (PR) curves

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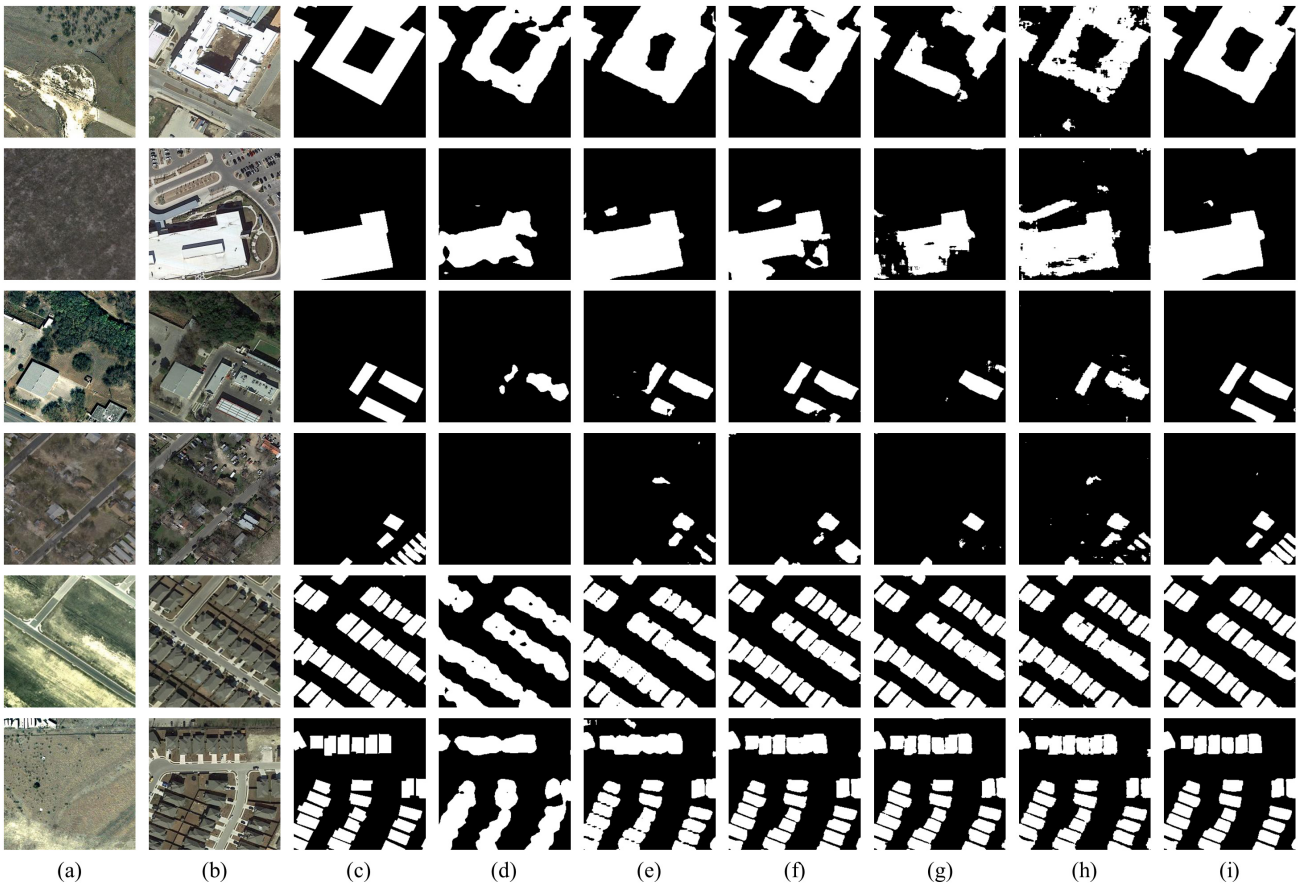


Fig. 8. Change detection examples of different methods on the LEVIR-CD test dataset. (a) T1 image. (b) T2 image. (c) Ground truth change map. (d) ChangeNet. (e) STANet. (f) UNet-LSTM. (g) MSOF. (h) FDCNN. (i) EGRCNN.

TABLE IV
GPU INFERENCE TIME OF THE PROPOSED METHOD AND THE COMPARISON METHODS

Methods	ChangeNet	STANet	UNet-LSTM	MSOF	FDCNN	EGRCNN
Time(ms)	19.94	22.19	17.55	11.37	8.24	23.36

of different methods on two datasets, as shown in Fig. 10. A point on the PR curve represents the precision and recall of the algorithm under a certain threshold. The entire PR curve is generated by sliding the threshold from low to high. We map the value of each pixel in the probability map from 0–1 to 0–255 and then use a sliding threshold from 0 to 255 to binarize the probability map. Because the PR curves of different methods are densely distributed, it is difficult to see the difference between these methods. We intercept part of the PR curves to make the difference between the curves more obvious. It can be seen from Fig. 10 that the PR curves of EGRCNN almost envelop the other curves, which shows that our method is superior to other methods on both datasets.

F. Time Consumption

The results of the GPU inference time of the proposed method and the comparison methods are shown in Table IV. The results in the table are obtained by calculating the average value of GPU inference time for all samples in the LEVIR-CD test dataset. All the experiments are implemented on a computing server equipped with an Intel Core i7-6850K CPU and an NVIDIA GeForce GTX 1080ti GPU. It can be observed that the FDCNN consumes minimal time, due to the lightweight training model. Besides, the MSOF has high computational efficiency because it directly analyzes the implicit relationship of the multitemporal images, without the process of predicting the deep features from the original image. In contrast, the UNet-LSTM, ChangeNet, STANet, and the proposed EGRCNN adopt the siamese network to extract deep features, and thus, these methods will take more running time. In order to improve the performance, the STANet uses residual blocks and attention mechanisms, while the EGRCNN method jointly learns both the building change detection and building edge estimation tasks, which further increases time consumption. It can be seen from experimental results that though time consumption by the proposed EGRCNN method has been increased slightly, the accuracy has been improved significantly.

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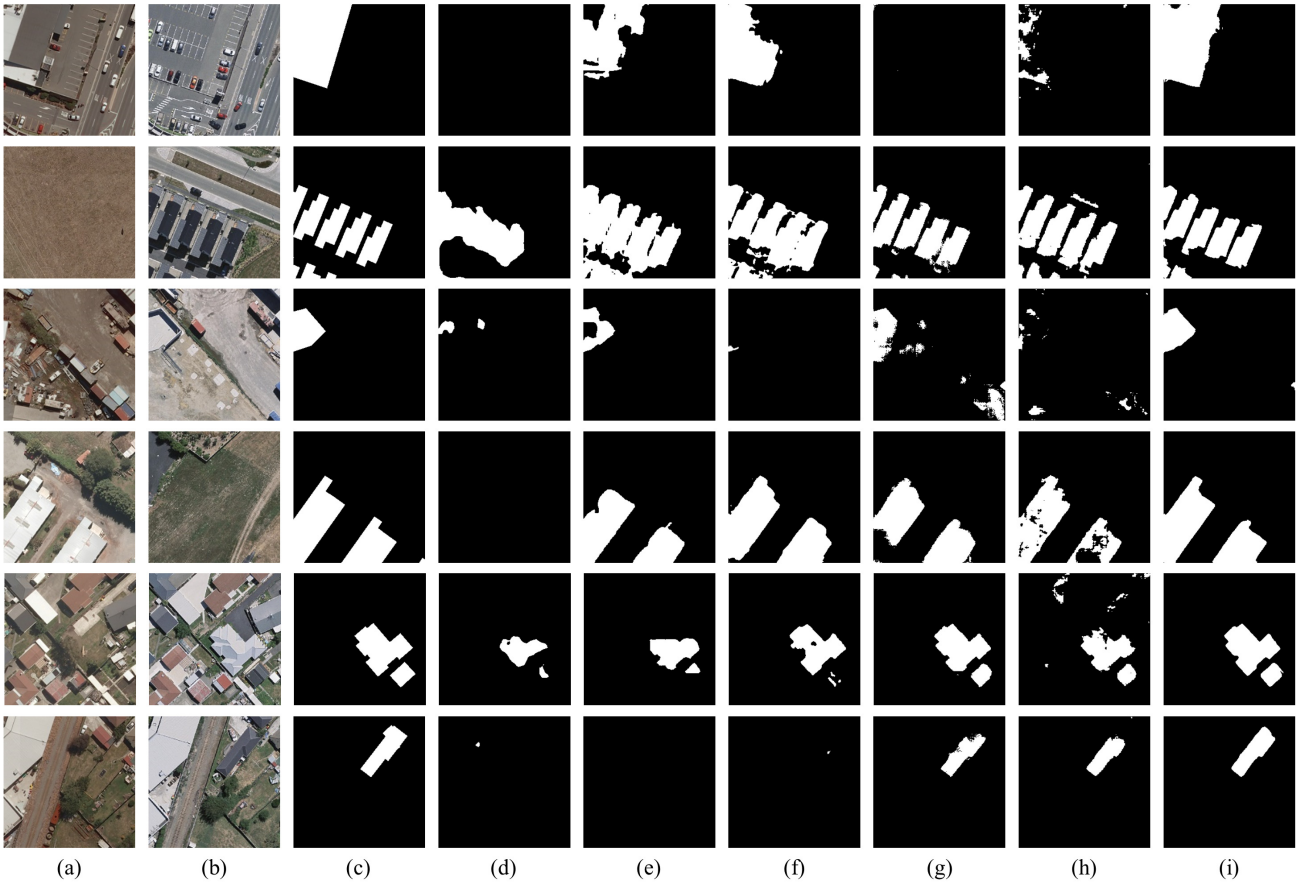


Fig. 9. Change detection examples of different methods on the WHU building change detection test dataset. (a) T1 image. (b) T2 image. (c) Ground truth change map. (d) ChangeNet. (e) STANet. (f) UNet-LSTM. (g) MSOF. (h) FDCNN. (i) EGRCNN.

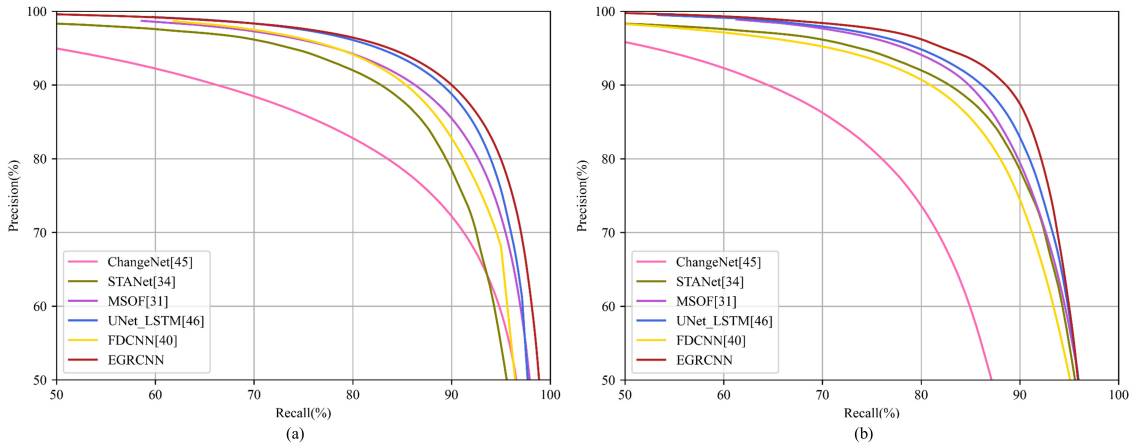


Fig. 10. PR curves of different methods on the two datasets. (a) LEVIR-CD dataset. (b) WHU building change detection dataset.

V. Conclusion

In this article, a novel method called EGRCNN is proposed for building change detection. The EGRCNN method mainly consists of three components: multilevel features extraction, difference information extraction, and edge-guided change map estimation. The proposed DAM adds an adaptive weighting mechanism on the basis of LSTM, which realizes the enhancement of difference information. In order to make the edges of buildings more accurate, the developed EGM integrates edge features into discriminative features. In addition, a deep edge-guided loss function is designed to learn changed building information and its edge at the same time. In the training process, the constraint of the loss function forces the final predicted changed building to preserve its original boundaries. Experimental results show that the proposed method achieves better performance on two public building change detection datasets, in terms of both numerical metrics and visual results.

The proposed change detection method based on DL can obtain satisfactory results. However, enough training samples are required to learn a good deep feature representation, which

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sometimes labor-costing and time-consuming. In practice, there are abundant unlabeled samples that are more easily to be obtained. Therefore, how to fully mine the association between labeled samples and unlabeled samples and develop a semi-supervised change detection method based on the proposed EGRCNN algorithm will be one focus of future work.

REFERENCES

[1] H. Luo, C. Liu, C. Wu, and X. Guo, "Urban change detection based on Dempster–Shafer theory for multitemporal very high-resolution imagery," *Remote Sens.*, vol. 10, no. 7, p. 980, 2018.

[2] K. Nemoto, R. Hamaguchi, M. Sato, A. Fujita, T. Imaizumi, and S. Hikosaka, "Building change detection via a combination of CNNs using only RGB aerial imageries," *Proc. SPIE*, vol. 10431, Oct. 2017, Art. no. 104310J.

[3] L. Li, C. Wang, H. Zhang, and B. Zhang, "Residual UNet for urban building change detection with Sentinel-1 SAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 1498–1501.

[4] J. Tian, S. Cui, and P. Reinartz, "Building change detection based on satellite stereo imagery and digital surface models," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 406–417, Jan. 2014.

[5] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. Mach. Process. Remotely Sensed Data Symp.*, 1980, pp. 1–12.

[6] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[7] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.

[8] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[9] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[10] L. Bruzzone and D. F. Prieto, "Automatic analysis of the difference image for unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1171–1182, May 2000.

[11] L. Bruzzone, R. Cossu, and G. Vernazza, "Detection of land-cover transitions by combining multidate classifiers," *Pattern Recognit. Lett.*, vol. 25, no. 13, pp. 1491–1500, 2004.

[12] J. S. Deng, K. Wang, Y. H. Deng, and G. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, 2008.

[13] A. Ertürk, M.-D. Iordache, and A. Plaza, "Sparse unmixing-based change detection for multitemporal hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 708–719, Feb. 2015.

[14] K. Ding, C. Huo, Y. Xu, Z. Zhong, and C. Pan, "Sparse hierarchical clustering for VHR image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 577–581, Mar. 2015.

[15] C. Wu, B. Du, X. Cui, and L. Zhang, "A post-classification change detection method based on iterative slow feature analysis and Bayesian soft fusion," *Remote Sens. Environ.*, vol. 199, pp. 241–255, Sep. 2017.

[16] X. Sun, P. Wang, C. Wang, Y. Liu, and K. Fu, "PBNet: Part-based convolutional neural network for complex composite object detection in remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 50–65, Mar. 2021.

[17] Y. Li, T. Lu, and S. Li, "Subpixel-pixel-superpixel-based multiview active learning for hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4976–4988, Jul. 2020.

[18] P. Duan, X. Kang, S. Li, P. Ghamisi, and J. A. Benediktsson, "Fusion of multiple edge-preserving operations for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10336–10349, Dec. 2019.

[19] Q. He, X. Sun, Z. Yan, and K. Fu, "DABNet: Deformable contextual and boundary-weighted network for cloud detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, early access, Jan. 5, 2021, doi: 10.1109/TGRS.2020.3045474.

[20] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[21] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[22] D. Hong *et al.*, "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, Apr. 2021.

[23] D. Hong *et al.*, "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[24] B. Bischke, P. Helber, J. Folz, D. Borth, and A. Dengel, "Multi-task learning for segmentation of building footprints with deep neural networks," in *Proc. 26th IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 1480–1484.

[25] Y. Xu, L. Wu, Z. Xie, and Z. Chen, "Building extraction in very high resolution remote sensing imagery using deep learning and guided filters," *Remote Sens.*, vol. 10, no. 1, p. 144, Jan. 2018.

[26] D. Marmanis, K. Schindler, J. D. Wegner, S. Galliani, M. Datcu, and U. Stilla, "Classification with an edge: Improving semantic image segmentation with boundary detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 135, pp. 158–172, Jan. 2018.

[27] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021, doi: 10.1109/JSTARS.2020.3037893.

[28] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[29] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on deep siamese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Oct. 2017.

[30] C. Zhang *et al.*, "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[31] D. Peng, Y. Zhang, and H. Guan, "End-to-end change detection for high resolution satellite images using improved UNet++," *Remote Sens.*, vol. 11, no. 11, p. 1382, Jun. 2019.

[32] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Cham, Switzerland: Springer, 2018, pp. 3–11.

[33] X. Peng, R. Zhong, Z. Li, and Q. Li, "Optical remote sensing image change detection based on attention mechanism and image difference," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 10, 2020, doi: 10.1109/TGRS.2020.3033009.

[34] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, 2020.

[35] H. Jiang, X. Hu, K. Li, J. Zhang, J. Gong, and M. Zhang, "PGA-SiamNet: Pyramid feature-based attention-guided siamese network for remote sensing orthoimagery building change detection," *Remote Sens.*, vol. 12, no. 3, p. 484, Feb. 2020.

[36] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral-spatial-temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[37] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[38] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

[39] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[40] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

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[41] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.
[42] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.
[43] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[44] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.
[45] A. Varghese, J. Gubbi, A. Ramaswamy, and P. Balamuralidhar, "ChangeNet: A deep learning architecture for visual change detection," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 1–16.
[46] M. Papadomanolaki, S. Verma, M. Vakalopoulou, S. Gupta, and K. Karantzalos, "Detecting urban changes with recurrent neural networks from multitemporal Sentinel-2 data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 214–217.



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